

CCU – SCU Communication Protocol

ITS System

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Version: 1.8

Sumith Electronics Private Limited

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Sumith Electronics Private Limited
Plot No 15A & B, SSRK Plaza, EC Complex ,
Kushaiguda, Hyderabad – 500062, T
Telangana, India

Sumith Electronics Private Limited

Plot No. 15 A & B , 3rd Floor ,
SSRKPlaza,EC Complex,
Kushaiguda Hyderabad-500062
Telangana,
India
e-Mail: contact@sumith.in

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About the Document

History

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1 Overview

2 Use cases requiring data transmission from OBU to CCU:

The following sections explain the packet details of the information expected to be sent by CCU to OBU, which is associated with each use case listed below.

Application	Information
OBU	On Board Unit (on Bus)
CCU	Close Control Unit (Server)
GPRS	General Packet Radio Service

3 Login Data(OBU-CCU)

Contents : Whenever OBU establishes the connection with backend server, Vehicle no, IMEI no, last valid location sent to server

Channel : GPRS

Communication Details : When GPRS is connected this data send to CCU

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$LGN
Packet Contents:			
2	Vehicle number where the device was installed	String	DL3CBM9821
3	15 Digit IMEI number	String	123456789012345
4	Version of the firmware used in	String	1.0.0
5	Version of the frame format protocol	String	1.0.1
6	Last location info saved at the device (gps fix, date (ddMMyy) time (HHmmss), latitude, latitude direction, longitude, longitude direction, speed)	String	1,220714,050656,28.758963,N,77.6277844,E,25
Packet Footer:			
7	32-bit Checksum Insures No error in transmission (optional)	Long (String)	2448041353
8	End Character (FIXED 1 char)	String	#
Sample packet :			
Example	\$LGN,AP36TS1234,864819050843380,11.0.20,1.0.2,1,120324,103232,17.47835333,N,78.5 7026167,E,0,18f6474f#		

Each field is separated by “,” (Comma) except first and last characters.

4 Location Data(OBU-CCU)

Contents : Location information including latitude, longitude, date, time, speed, status of units.

Channel : GPRS

CommunicationDetails : When GPRS is connected location, data is sent to server by specified interval

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	LOC
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Firmware Version	String(n)	1.0.0
5	Packet Type (NR=Normal EA =Emergency alert TA = Tamper alert HP = Health packet IN = Ignition On IF = Ignition Off BD = Main Battery disconnect (If device is disconnected from vehicle battery and running on its internal battery) BR = Battery reconnect BL = Battery low (If device internal battery had fallen below a defined threshold, indicating that device need to get a recharge) BC = (Low battery removed Indicate that vehicle internal battery is charged again) GO (– GPS box opened Message would be generated indicating GPS box opened) optional HB (Harsh Break) HA (Harsh Acceleration) PC (parameter change)	NR ix) Alert for over the air parameter change Alerts for any parameter changed over the air/manually. Shall include the name/value of parameter changed and source of command. X	NR

	RT(Rash Turning)		
6	Alert ID: 1 -Location Update 2- Location Update(history) 3- Alert - Disconnect from main battery 4- Alert - Low battery 5- Alert- Low battery removed 6- Alert - Connect back to main battery 7- Alert-Ignition ON 8- Alert - Ignition OFF 9- Alert - GPS box opened (Optional) 10- Alert - Emergency state ON 11- Alert - emergency State OFF 12- Alert Over the air parameter change 13- Harsh Braking 14- Harsh Acceleration 15- Rash Turning 16- Device Tempered	1	1
7	Packet Status L=Live or H= History	String(1)	L
8	15 Digit IMEI number	String	123456789012345
9	Vehicle number where the device was installed	String	DL3CBM9821
10	GPS Fix (1 = GPS fix OR 0 = GPS invalid)	String(1)	1
11	Date value (ddmmYYYY)	String(6)	27072024
12	Time in UTC format (hhmmss)	String(6)	102015
13	Latitude value in decimal degrees (with minimum 6 decimal places)	double(String)	17.7823411
14	Latitude Direction (For example, N=North;S=South)	String(1)	N
15	Longitude value in decimal degrees (with minimum 6 decimal places)	double(string)	78.3123123
16	Longitude Direction (For example, E=East;W=West)	String(1)	E
17	Speed in km/h	String(n)	25.1
18	Heading Course over ground in degrees	String(n)	310.56
19	No of satellites	String	15
20	Altitude of the device in meters	String (n)	183.5
21	PDOP Horizontal dilution of precision	String(n)	4
22	HDOP Horizontal dilution of precision	String(n)	4
23	Network Operator Name	String	Airtel

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24	Ignition 1= Ign On ; 0 = Ign Off	String(1)	1
25	MainPower Status: 0 = Vehicle BatteryDisconnected 1 = Vehicle battery power reconnected	String(1)	0
26	Main Input voltage	String	23.72
27	Internal battery Voltage	String	8.18
28	EmergencyStatus 1= 'On'; 0 = 'Off'	String(1)	0
29	Tamper Alert (C =Cover Closed; O = Cover Open)	String(1)	C
30	GSM Signal Strength Value ranging from0 – 31	String(1)	22
31	MCC Mobile country code	String(n)	404
32	MNC Mobile network code	String(n)	10
33	LAC Location area code	Short (String)	(214) (0x00D6)
34	CellID GSM Cell ID	Int (String)	53181 (0xCFBD)
35	NMR(Neighboring 4 cell ID along with their LAC and signal strength)(separated with #)	(String)	845#300#067# 234#289#076# 342#211#0789 #243#453#098
36	Digital Input Status1234 (0=Off; 1=On)	String	0000
37	Digital Output Status12 (0=Off; 1=On)	String	00
38	Frame Number Sequence Numberof the messages (000001 to 999999)	Long(String)	000001
39	On Parameter change packet Only. If OTA through server ipaddress and port if OTA thorough phone phonenumber	String	9898786754
40	On Parameter change packet Only. Parameter changed values e.g:ip,apn etc	String	jionet
Packet Footer:			
41	32 bit ChecksumInsures No error intransmission(optional)	Long(String)	
42	End Character	String	#
GPRS Sample packet :			
Example	\$LOC,3339,11.0.20,NR,1,L,864819050843380,AP36TS1234,1,12032024,050402,17.47834833,N,78.57026333,E,0.0,191.0,18,562.245,1.29,0.65,jionet,1,1,22.77,7.61,0,C,18,404,73,A51,AABF,0#0#0#0#0#0#0#0#0#0#0,0000,00,000794,3dde40e6#		

5 Health Status Data(OBU-CCU)

Frequency : Configurable interval

Contents : Health Status of OBU . When GPRS is connected, send health status of OBUfor every configurable interval of time to Server.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String(1)	\$HLT
2	OBU ID	String(15)	000009
3	mDVR Name	String(3)	Dik3339
4	Vendor ID (FIXED 6 chars)	String(6)	SU3339
5	Datedddmmyyyyhhmmss and time	String	20122024123314
6	primaryIP Address (String)	String	120.33.34.1
7	SecondaryIP Address (String)	String	120.12.2.1
8	FirmwareVersion (string)	String	1.0.0
9	Protocol Version(String)	String	1.0.1
10	IMEI number	String	123456789012345
11	Storage 1 status (0 Failure/1 Working)	String	1
12	Storage 1 memorystatus (0 Failure/1 Working)	String	1
13	Storage 2 status (0 Failure/1 Working)	String	1
14	Storage 2 memorystatus (0 Failure/1 Working)	String	1
15	Camera 1 recording status(0 Failure/1 recording)	String	1
16	Camera 2 recording status(0 Failure/1 recording)	String	1
17	Camera 3 recording status(0 Failure/1 recording)	String	1
18	Camera 4 recording status(0 Failure/1 recording)	String	1
19	Camera 5 recording status(0 Failure/1 recording)	String	0
20	Camera 6 recording status(0 Failure/1 recording)	String	0
21	Camera 7 recording status(0 Failure/1 recording)	String	0
22	Camera 8 recording status(0 Failure/1 recording)	String	0
23	Microphone 1 recording status(0 Failure/1 recording)	String	0
24	Microphone 2 recording status(0 Failure/1 recording)	String	0
25	Microphone 3 recording status(0 Failure/1 recording)	String	0
26	Microphone 4 recording status(0 Failure/1 recording)	String	0

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27	Microphone 5 recording status(0Failure/1 recording)	String	0
28	Microphone 6 recording status(0Failure/1 recording)	String	0
29	Microphone 7 recording status(0Failure/1 recording)	String	0
30	Microphone 8 recording status(0 Failure/1 recording)	String	0
31	Ignitionstatus (0 Off/1On)	String	0
32	Emergencybuttons status (0 Off/1 On)	String	0
33	Battery Percentage(Indicates theInternal batterycharge percentage)	String	80
34	Low Battery threshold value(Indicates value on which low battery alert generated in percentage)	String	7.0
35	Memory percentage(Indicates flash memorypercentage used)	String	50
36	Data updatarate when ignition on	String	10
37	Data update rate when ignition off	String	15
38	Digital I/O status	String	0000
39	Analog I/O status(1-connected,0-disconnected)	String	0
Packet Footer:			
41	32 bit ChecksumInsuresNo error in transmission(optional)	Long(String)	232246 4851
42	End Character	String	#
HealthStatus DataSample packet :			
Example	\$HLT,000009,Dik3339,3339,12032024093947,49.207.2.123,122.168.23.1,11.0.20,1.0.2,8 64819050843380,1,1,1,1,1,1,1,1,1,1,0,0,0,0,0,0,0,1,0,40,7.0,23.66,5,15,0000,0,4be925ca#		

6 Alert on Emergency button press(OBU-CCU)

Frequency : On press of Emergency Button

Contents : Emergency Message on Press of Emergency button

Channel : GPRS

Communication Details : When GPRS is connected, Send emergency message to dual IP Address. Until OBU receives STP message from server send the packet with configurable interval of time. Once OBU receives STP message send STOP_ACK message to server

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String(1)	\$
2	Packet Header	String (3)	EPB
3	Message type(FIXED 3 chars)	String(3)	EMR/SEM
4	IMEI(Vehicleid)	String	123456789012345
5	Packet Type	String	NM / SP
6	Date and time(ddmmyyyyhhmmss format)	String	20200728122035
7	GPS valid (A-Valid, V-Invalid)	String	A
8	Latitude in decimal degrees - dd.mmmmmm format	String	17.1234561231
9	Latitude Direction N – North; S – South	String	N/S
10	Longitude in decimal degrees - dd.mmmmmm format	String	78.123456781
11	Longitude Direction E – East; W – West	String	E/W
12	Altitude in 'm'	String	235.123312312
13	Speed in km/h	String	38.123
14	Distance calculated from previous GPS data	String	123.31
15	Provider (G – FineGPS N – CoarseGPS or data from the network)	String	G/N
16	Vehicle Registration number	String	AP361234
17	Reply Number (The mobile number to which test response need to be sent)	String	9876543210
Packet Footer:			
18	End Of Packet	String	#
19	The 32-bit checksum of all the characters from the header up to the CRC field	String	
Sample Packet			
Example	\$EPB,EMR,864819050843380,NM,12032024112442,A,17.478336666,N,78.570258333,E,578.95600000,0.0000,0.0000,G,AP36TS1234,9898787654,#7ed424dd		

7 Send Firmware Version(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get Firmware Version** Command, OBU has to send this packet

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(3)	101
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet details			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Firmware Version	String	1.0.0
Packet Footer			
8	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample Packet:			
Example	\$101,3339,AP36TS1234,864819050843380,240312115645,11.0.20,70bf1fd4#		

8 Send Protocol Version: (OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get Protocol Version** Command,OBU has to send this packet.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(3)	102
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
Packet Footer:			
7	Protocol Version	String	1.0.0
Packet Footer:			
8	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample Packet:			
Example	\$102,3339,AP36TS1234,864819050843380,240312115659,1.0.2,75f2b009#		

9 Send Mac Address(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get MAC Address** Command, OBU has to send this packet

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	103
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Mac Address	String	248.89.113.5.7. 103
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample Packet:			
Example	\$103,3339,AP36TS1234,864819050843380,240312115715,d8:3a:dd:83:9e:41,9e770805#		

10 Send Primary IP Address(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get Primary IP Address** Command, OBU has to send this packet

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	104
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If getipaddress & port came through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	IP Address	String	222.168.83.2
9	PORT NO (short)	String	9999
Packet Footer:			
10	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
11	End of Packet	String	#
Sample packet :			
Example	\$104,3339,AP36TS1234,864819050843380,240312115758,49.207.2.123,2255,49.207.2.123,2255,49014367#		

11 Send Secondary IP Address(OBU-CCU)

Frequency : Event Based

Contents :Whenever OBU receives Get Secondary IP Address Command,OBU has to send this packet.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	105
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If getipaddress&port change through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	IP Address	String	222.168.83.2
9	PORT NO (short)	String	9998
Packet Footer:			
10	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
11	End of Packet	String	#
Sample packet :			
Example	\$105,3339,AP36TS1234,864819050843380,240312115810,49.207.2.123,2255,49.207.2.123,2288,2472da5c#		

12 Send IMEI Number(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get IMEI Number** Command, OBU has to send this packet.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	106
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If getimei came through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	IMEI Number	String	123456789123456
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$106,3339,AP36TS1234,864819050843380,240312115833,49.207.2.123,2255,864819050843380,f6a7c331#		

13 Send Vehicle Registration Number(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get Registration Number** Command, OBU has to send this packet

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	107
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If get vehicle Number came through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	Vehicle Registration Number	String	123456789123456
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$107,3339,AP36TS1234,864819050843380,240312115736,49.207.2.123,2255,AP36TS1234,70bd8311#		

14 Send Ignition Mode(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get Ignition Mode** Command, OBU has to send this packet

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	108
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenumber where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If get ignition mode came through Server/SMS Server-ip address and port SMS-phonenum	String	9898989898
8	Emergency button Mode 1 Enable 0 Disable ☐	String	1
9	SMS ☐ Mode 1 Enable 0 Disable ☐	String	1
10	Emergency Transmission Interval(seconds) (1 to 255)	String	5
11	Sleep mode/shutdown Mode Sleep mode 0 ☐ Shutdown mode 1	String	1
12	Delay to enter into sleep/shutdown mode(min) (1 to 255) ☐	String	15
13	Frequency in Sleep mode(seconds) (1 to 255)	String	30
14	Frequency in normal mode(seconds) (1 to 255)	String	10
15	Phonenum	String	9898989898
Packet Footer:			
14	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
15	End of Packet	String	#
Sample packet :			
Example	\$108,3339,AP36TS1234,864819050843380,240312115747,49.207.2.123,2255,1,0,10,0,10,15,5,1,76f1efd3#		

15 Common Acknowledgement to required packets (OBU to CCU)

Frequency : Event based

Contents : Every time OBU receives packet from CCU, it sends back acknowledgement

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	ACK
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	1 success 0 Failure	String	1
8	Message type of received Packet Sample: Adhoc: 52 Main Stream Resolution: 101 Overspeed buzzer: 006 Clear all: 007	String	60
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$ACK,3339,AP36TS1234,864819050843380,240312141422,1,082,75af674c#		

16 Panic message (OBU to CCU):

Frequency : Event based; everytime driver presses the“Panic” button on the OBU

Contents : PANIC event

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character(FIXED1 char)	String (1)	\$
2	Header (FIXED 3chars)	String (3)	002
3	Vendor ID (FIXED 6chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Panic ID, 01 – Terrorist Attack 02 – Bus Hijacked 03 – Met Severe Accident 04 – Bus Caught Fire 05 – Drowning in Water	String	1
8	Latitude (00.000000000)	String	17.566555
9	Longitude (00.000000000)	String	78.566895
Packet Footer			
10	32-bit Checksum Insures No error in transmission(optional)	Long(String)	
11	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$002,SU3339,000009,123456789123456,201124135236,1,17.47792333,78.57027833,38 71289600#		

17 Current Route Start (OBU to CCC):

Frequency : Event based, every time driver starts a new route

Contents : Route ID selected by the driver (OBU to CCC)

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	050
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Current route id	String	20P
9	Expected Completion Time, Format=HHMM	String	1220
10	Duty Id, Format= 'X1Y2' or 'Y2' (Maximum 16 character)	String	0000000000123123
11	Trip Id, Format= '01' or 'FF' or '0F' (Maximum 2)	String	1
Packet Footer:			
12	32 bit Checksum Insures No error in transmission	Long (String)	
13	End Byte, Value= "#"		#
Sample packet :			
Example	\$050,SU3339,DL3CBM9821,123456789123456,201120121141,1,201120120933,39691 69766#		

18 Cancelled Route:

Frequency : Event based

Contents : Whenever Cancel route is selected by the driver

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	054
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	Cancelled Route ID, Format = ABCDEF (Maximum 6 character) Alphanumeric	String	20P
8	DutyId, Format = 'X1Y2' or 'Y2' (Maximum	String	0000000000123123
9	Trip Id, Format = 1 to 100	String	1
Packet Footer:			
10	32-bit Checksum Insures No error in	Long (String)	
11	End Byte, Value = "#"	String	#
Sample packet :			
Example	\$054,SU3339,000009,123456789123456,201124140018,112GU,1234,38,1919020198#		

19 GPRS packet on Approaching Stop (OBU to CCC):

Frequency : Event based, every time when vehicle approaches to any stop.

Contents : Stop id, route id and duty id for the respective stop.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	062
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	000000000123123
8	Route id	String	20P
9	Stop id	String	SECUNDE R ABAD
10	Reserved byte	String	0
11	Trip Id, Format= '01' or 'FF' or '0F' (Maximum 2 character)	String	1
Packet Footer:			
12	32 bit Checksum Insures No error in transmission	Long (String)	
13	End Byte, Value= "#"	String	#
Sample packet :			
Example	\$062,SU3339,000009,123456789123456,201123160815,1234,112ED,IFFCO CHOWK,,33,4122297067#		

20 GPRS packet on Reach Stop (OBU to CCC):

Frequency : Event based, every time when vehicle reach to any stop.

Contents : Stop id, route id and duty id for the respective stop.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	063
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234
8	Route id	String	20p
9	Stop id	String	SECUNDRABA
10	Reserved byte	String	
11	Trip Id, Format = '01' or 'FF' or '0F' (Maximum 2 character)	String	01
Packet Footer:			
12	32 bit Checksum Insures No error in transmission	Long (String)	
13	End Byte, Value=" #"	String	#
Sample packet :			
Example	\$063,SU3339,000009,123456789123456,201123160654,1234,112ED,IFFCOCHOWK,,33,2555736085#		

21 GPRS packet on Depart Stop (OBU to CCC):

Frequency : Event based, every time when vehicle departs from any stop.

Contents : Stop id, route id and dutyid for the respective stop.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	064
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Dutyid	String	1234
8	Route id	String	112ED
9	Stop id	String	IFFCO CHOWK
10	Reserved byte	String	
11	Trip Id, Format= '01' or 'FF' or '0F' (Maximum 2)	String	33
Packet Footer:			
12	32 bit Checksum Insures No error in transmission	Long (String)	3379106681
13	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$064,SU3339,000009,123456789123456,201123160953,1234,112ED,IFFCO CHOWK,,33,3379106681#		

22 GPRS packet on Skip Stop (OBU to CCC):

Frequency : Event based, every time when vehicle skips any stop.

Contents : Stop id, route id and duty id for the respective stop.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	065
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
8	Duty id	String	1234
9	Route id	String	112ED
10	Stop id	String	IFFCO CHOWK
11	Reserved byte	String	
12	Trip Id, Format= '01' or 'FF' or '0F' (Maximum 2 character)	String	1
Packet Footer:			
13	32-bit Checksum Insures No error in transmission	Long(String)	3379106681
14	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$065, SU3339,000009,123456789123456,201123160953,1234,112ED,IFFCO CHOWK,,33,3379106681#		

23 GPRS packet on Route End(OBU to CCC):

Frequency : Event based, every time on route end.

Contents : Duty id and route id for the respective route.

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	066
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234
8	Route id	String	112ED
9	Trip Id, Format= '01' or 'FF' or '0F' (Maximum 2)	String	33
Packet Footer:			
10	32 bit Checksum Insures No error in transmission	Long (String)	3732040895
11	End Byte, Value= "#"	String	#
Sample packet :			
Example	\$066,SU3339,000009,123456789123456,201123155700,1234,112ED,33,3732040895#		

24 GPRS packet on Duty Start (OBU to CCU):

Frequency: Event based, every time when duty start.

Contents : Dutyid for the respective duty.

Channel : GPRS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	067
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234567891
8	Reserved String	String	
Packet Footer			
10	32 bit ChecksumInsures No error intransmission	Long(String)	2690877089
11	End Byte, Value="#"	String	#
Sample packet :			
Example	\$067,SU3339,000009,123456789123456,201123160457,1234,,2690877089#		

25 GPRS Packet on Duty End (OBU to CCU) :

Frequency : Event based, every time when duty ends.

Contents : Duty id for the respective duty.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	068
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234
8	Reserved String	String	
Packet Footer:			
9	32 bit Checksum Insures No error in transmission	Long(String)	
10	End Byte, Value="#"	String	#
Sample packet :			
Example	\$068,SU3339,000009,123456789123456,201123160206,1234,,3609845968#		

26 GPRS Packet on Over Speed Start Message (OBU to CCU):

Frequency : Event based

Contents : Everytimedriver surpasses configured upper speed limit

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	071
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEInumber	String	123456789012345
6	DateandTime, Format=YYMMDDHHMMSS	String	200728121324
7	Upper speed limit	String	30
8	Current speed	String	45
9	Latitude (00.00000000 0)	String	17.233234
10	Longitude(00.000000000)	String	78.412312
Packet Footer:			
11	32 bit ChecksumInsures No error in transmission	Long(String)	
12	EndByte, Value="#"	String	#
Sample packet :			
Example	\$071,SU3339,000001,863958048416379,210102173309,10,11,16.83432333,78.15490333,1371366855#		

27 Over Speed End Message (OBU to CCU):

Frequency : Event based,

Contents : every time driver surpass configured upper speed limit and return to upper speed limit.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	072
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Upper speed limit (in decimal)	String	30
8	Current speed (in decimal)	String	28
9	Max speed attained (in decimal)	String	40
10	Duration (unit seconds-in decimal)	String	10
11	Latitude (00.000000000)	String	17.23123
12	Longitude (00.000000000)	String	78.123123
Packet Footer:			
13	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
14	End Byte, Value="#"	String	#
Sample packet :			
Example	\$072,SU3339,000001,863958048416379,210102175339,10,9,11,0,16.87611833,78.16604 83,2118132307#		

28 GPRS Packet on Vehicle Stoppage (OBU to CCU):

Frequency : Event based, every time when vehicle stops for more than configured time.

Contents : Vehicle stoppage limit & duration

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	073
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Vehicle Stoppage Duration limit	String	15
8	Duration	String	5
9	Latitude (00.000000000)	String	17.123123
10	Longitude (00.000000000)	String	78.123123
Packet Footer:			
11	32 bit Checksum Insures No error in	Long(String)	
12	End Byte, Value= “#”	String	#
Sample packet :			
Example	\$073, SU3339DL3CBM9821,863958048416379,210102175339,15,5,16.87611833,78.166048 33,2118132307#		

29 GPRS packet on Reset (OBU to CCU):

Frequency : Event based, everytime when OBU receives Reset Command then OBU sends response to Server.

Contents : Reset ON event

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	081
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=	String	200728121324
7	Reserved byte	String	
Packet Footer:			
8	32 bit Checksum Insures No	Long(String)	
9	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$081,SU3339,AP36TS1234,863958048416379,201226151612,,563526300#		

30 Driver Halt Justification (OBU to CCU):

Frequency : Event based, everytime when Driver halts at non bus stop.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	084
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	ID (Justification ID)	String	5
Packet Footer:			
8	32 bit Checksum Insures No error in transmission	Long (String)	
9	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$084,SU3339,DL3CBM9821,863958048416379,201226151612,5,563526300#		

31 Schedule Request(OBU-CCU)

Frequency : Every time OBU is switched On during power on/reset, OBU will send Schedule request packet, with current Duty ID, if current Duty ID is not present then OBU will send ID as zero, server should send response packet either success or failure. If schedule ID doesn't match or incorrect, then Server should send permanent schedule packet to OBU.

Contents : Schedule Request

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1)	String	\$
2	Header (FIXED 3chars)	String (3)	085
3	Vendor ID (FIXED 6)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	DutyID	String	1234
Packet Footer:			
8	32-bit Checksum	Long(String)	
9	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$085,SU3339,000009,123456789123456,201124140921,1234,1041193168#		

32 GPRS Packet on Special Message(OBU-CCU)

Frequency : Event based, everytime when special message is selected on OBU.

Contents : Special Message Selected.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	086
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Message ID, 01 – Special Bus On Route 02 – Bus Break down 03 – Bus On Trial 04 – Bus Off Route Bus 05 – Operation change from Feeder route to BRTS corridor	String	1
Packet Footer:			
8	32-bit Checksum Insures No error in transmission	Long(String)	
9	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$086,SU3339,000009,123456789123456,201123162004,1,2954338138#		

33 GPRS Packet on Ping Connection Check Packet(OBU-CCU)

Frequency : Event based, to check GPRS connection is available or not.

Contents : Connection check just as ping

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	090
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	
Packet Footer:			
8	32 bit Checksum Insures No error in	Long(String)	
9	End Byte, Value="#"	String	#
Sample packet :			
Example	\$090,SU3339,000001,863958048416379,210102174133,0,2305114412#		

34 GPRS Packet on OBU Started (OBU to CCU):

Frequency : Event based, everytime when OBU is turned on/restarted.

Contents : OBU ON event

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	094
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	0 ☐ Normal Power on 1 ☐ Watchdog Reset 2 ☐ Low Power reset	String	1
Packet Footer:			
8	32 bit Checksum Insures No error in	Long(String)	
9	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$094,SU3339,000009,123456789123456,201124144239,3,1447636470#		

35 Vehicle Electrical Systems ITS DTCs:

Frequency : Every time a DTC is encountered, it is collected for a user defined interval and then send to server

Contents : DTC Code, which can uniquely identify the DTCs as well as the equipment ID initiating the DTC.

Channel : GPRS

Configuration : Launch the OBITS Configuration Manager (OBITS Utility) go to the Others - >GPRS and enable the DTC data transmission.

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	022
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Device ID, as follows 00: LED-Front 01: LED-Side 02: LED-Rear 03: LED-Inner 10: SCU 20: GPS 30: USB	String	1
8	DTC Count	String	2
9	DTC ID 2 bytes	String	
10	DTC value 1 byte (00 inactive , 01 active)	String	
:	:	String	
:	:	String	
Packet Footer:			
n-1	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
N	End Byte, Value	String	#
Sample packet :			
Example	\$022,SU3339,TS99S0420,864337059041138,220621155541,10,0,10,0,122582280#		

36 Vehicle Electrical Systems (LED) ITS PIDs:

Frequency : Once, at SCU start up

Contents : LED PIDs and Values

Channel : GPRS

Configuration : Launch the OBITS Configuration Manager (OBITS Utility) go to the Others - >GPRS and enable the data transmission for LED PID

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	021
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Device ID, as follows 00: LED-Front 01: LED-Side 02: LED-Rear 04: LED-Inner	String	
8	Parameter Identifier/PID Requested for	String	
9	Parameter Value/PID Value	String	
Packet Footer:			
10	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
11	End Byte, Value	String	#
Sample packet :			
Example	\$021,SU3339,TS99S0420,864337059041138,220621155541,01,0,0,122582280#		

Note: Only 1 PID is transmitted at a time.

37 Vehicle Electrical Systems (SCU) ITS PIDs:

Frequency : Once, at SCU start up

Contents : SCU PIDs and Values

Channel : GPRS

Configuration : Launch the OBITS Configuration Manager (OBITS Utility) go to the Others - >GPRS and enable the data transmission for SCUPID

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	023
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
8	Parameter Identifier (PID)	String	
9	value of the PID	String	
Packet Footer:			
10	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
11	End Byte, Value	String	#
Sample packet :			
Example	\$023,SU3339,TS99S0420,864337059041138,220621155541,102,1.0,969249816#		

Note: Only 1 PID is transmitted at a time.

38 Pre-selected CAN Parameters:

Frequency : At a set frequency

Contents : Value of therequested CAN Parameter

Channel : GPRS

Configuration : Launch the OBITS Configuration Manager (OBITS Utility) go to the Others–>GPRS and enable thedata transmission for CANPID

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	010
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Dateand Time, Format= YYMMDDHHMMSS	String	200728121324
7	CAN length	String	
8	PGN No	String	
9	SPN No	String	
10	Value	String	
11	The sequence of the PGN-SPNs whose values need transmitted to CCU, is expected to be configured atSCU. The SCU will then transmit the CAN Parameter values as a sequential byte stream. The length ofeach CAN parameter will follow the J1939 specifications.	String	
Packet Footer:			
12	32 bit ChecksumInsures No error intransmission (optional)	Long(String)	
13	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$010,SU3339,TS99S0420,864337059041138,220621155541,22,3,4,12,969249816#		

39 CAN Alarms:

Frequency : Every time the pre-configured alarm is activated / deactivated, it is collected for a user defined interval and then sent to server

Contents : AlarmID, along with Parameter ID, Current Value, Alarm Type and Alarm Threshold

Channel : GPRS Launch the OBITS Configuration Manager (OBITS Utility) go to the Others-> GPRS and enable the data transmission for CAN Alarms

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	011
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	Alarm ID, unique unsigned integer that identifies the Alarm (An Alarm is a configurable item, which identifies the PGN-SPN, Alarm Type, Threshold value and is defined at the SCU)	String	
8	Alarm Status 01: Active 00: Inactive	String	1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$011,SU3339,TS99S0420,864337059041138,220621155541,102,01,969249816#		

40 Display Board Connectivity(OBU-CCU)

Frequency : Everytimethe Display Board is Connected/Disconnected, it is collected for a user defined interval and then send to server

Contents : Displayboard no and status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	096
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format= YYMMDDHHMMSS	String	200728121324
7	DisplayBoardNo Front :1 Side :2 Rear:3 Internal:4 □ □	String	1
8	Status (1 Connected, 0 Disconnected)	String	1
Packet Footer:			
9	32 bit ChecksumInsures No error intransmission (optional)	Long(String)	
10	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$096,SU3339,000009,123456789123456,201124144012,1,0,113018582#		

41 Video Connectivity(OBU-CCU)

Frequency : Every time the Video is activated / deactivated, it is collected for a user defined interval and then send to server

Contents : Video no and status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	093
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Video No (1,2,3,4)	String	1
8	Status (1 Connected, 0 Disconnected)	String	1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	2757815055
10	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$093,SU3339,000009,123456789123456,201124144012,1,0,2757815055#		

42 Door Open:

Frequency : Every time when Bus Door is Opened, then from CAN data if this parameter is configured, then this packet is sent to server

Contents : Status of Door open with location

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	097
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenumber wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format=YYMMDDHHMMSS	String	200728121324
7	Door No (1/2)	String	1
8	Speed	String	34
9	Latitude (00.000000000) 12 characters length	String	17.123123
10	Longitude(00.000000000) 12 characters length	String	78.123123
Packet Footer:			
11	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
12	End Byte, Value	String	#
Sample packet :			
Example	\$097,SU3339,DL3CBM9821,123456789123456,201124144012,1,0,17.123123,78.123123,2757815055#		

43 Door Close:

Frequency : Every time when Bus Door is closed, then from CAN data if this parameter is configured, then this packet is send to server

Contents : Status of Door opens with location

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	098
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Door No (1/2)	String	1
8	Speed	String	56
9	Latitude (00.000000000)	String	17.1231231321
10	Longitude (00.000000000)	String	78.1312312123
Packet Footer:			
11	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
12	End Byte, Value	String	#
Sample packet :			
Example	\$098,SU3339,DL3CBM9821,123456789123456,201124144012,1,0,17.123123,78.123123,2757815055#		

44 Response Packet from Server for Emergency Alert(CCUCU-OBUCU)

Frequency : On press of Emergency Button Response fromServer
on receiving help.

Contents : Response Message on Press of Emergency button

Channel : GPRS

CommunicationDetails : Receives response emergency messagefromserver
For emergency packet.

Packet details :

Sno	Contents	No ofBytes	Sample Data
1	Start Character (FIXED 1char)	String	\$
2	Header (FIXED 3 chars)	String (3)	EMR
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEInumber	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Response (Success - → 1/Failure → 0	String	1
Packet Footer:			
8	32-bit ChecksumInsures No error in transmission (optional)	Long(String)	
9	End ofPacket	String	#
Sample packet :			
Example	\$EMR, SU3339,DL3CBM9821,123456789123456,201124142256,1,1088473292#		

45 Camera Main Stream Video Resolution:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to change the main stream video resolution
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	101
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	Camera no (CAM1 1)	String	1
8	Video Resolution High 1 ☐ Above Normal 2 ☐ Medium 3 ☐ Medium 4 ☐ Low 5 ☐	String	1
Packet Footer			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$101, SU3339, DL3CBM9821, 1234567890123456, 20201120123531, 1, 5, 4113276741#		

46 Camera Main Stream image quality:

Frequency	: Whenever OBU receives command fromCCU
Contents	: Commandfrom CCUto change the main stream Image Quality
Channel	: GPRS
Packet details	:

Sno	Contents	No ofBytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	102
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevicewas installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image quality Range 1 to 6	String	1
Packet Footer			
9	32 bit Checksum Insures No error intransmission (optional)	Long(String)	
10	End ofPacket	String	#
Sample packet :			
Example	\$102,SU3339,DL3CBM9821,123456789123456,20201120125645,7,2,1520861599#		

47 Camera Main Stream Bitrate:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to change the main stream Bitrate
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	103
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Bitrate (unsigned integer)	String	512
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$103,SU3339,DL3CBM9821,123456789123456,20201120125910,1,1536,41131 73932#		

48 Camera Main Stream Bit Rate Type:

Frequency : Whenever OBU receives command fromCCU

Contents : Command fromCCUto change the main streamBit rate Type

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	104
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Bitrate Type (VBR 1 CBR 2) ☐	String	1
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$104,SU3339,DL3CBM9821,123456789123456,20201120130209,7,2,876474140#		

49 Camera Main Stream Max Bit rate :

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the main stream Max Bit rate

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	105
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Max Bitrate (256,512,1024,1536,2048)	String	512
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$105,SU3339,DL3CBM9821,123456789123456,20201120130556,7,1536,3334 9944 63#		

50 Camera Main Stream Frame Rate:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the main stream Frame rate

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	106
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Frame rate 1 to 25	String	24
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$106,SU3339,DL3CBM9821,123456789123456,20201120135405,7,5,3292631880#		

51 Camera Main Stream Frame Interval :

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the main stream Frame Interval

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	107
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Frame Interval (24 to 100)	String	50
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$107,SU3339,DL3CBM9821,123456789123456,20201120135554,1,45,1776435 092 #		

52 Camera Sub-Stream Video Resolution:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the sub stream video resolution

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	108
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Resolution High 1 Above Normal 2 Medium 3 Medium 4	String	1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$108,SU3339,DL3CBM9821,123456789123456,20201120135741,7,5,578143759#		

53 Camera Sub-Stream image quality:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to change the sub stream Image Quality
Channel	: GPRS
Packet details	:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	109
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image quality 1 to 6	String	1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$109,SU3339,DL3CBM9821,123456789123456,20201120140025,7,2,368624603#		

54 Camera Sub-Stream Bit rate :

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the Sub stream Bitrate

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	110
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Bit rate (unsigned integer) (256,512,1024,1536,2048)	String	512
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$110,SU3339,DL3CBM9821,123456789123456,20201120140236,7,1024,12140 757 55#		

55 Camera Sub-Stream Bit rate Type:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the sub stream Bit rate Type

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	111
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
Packet Contents			
7	Camera no	String	1
8	Video Image Bitrate Type (VBR 1 CBR 2)	String	1
Packet Footer: ☐			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$111,SU3339,DL3CBM9821,123456789123456,20201120140413,7,2,2700786 103#		

56 Camera Sub-Stream Max Bit rate :

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the sub stream Max Bit rate

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	112
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Max Bitrate (unsigned integer) (256,512,1024,1536,2048)	String	512 (0x02,0x00)
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$112,SU3339,DL3CBM9821,123456789123456,20201120140605,7,2048,1344 8605 19#		

57 Camera Sub-Stream Frame Rate :

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the sub stream Frame rate

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	113
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Image Frame rate 1 to 25	String	24
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$113,SU3339,DL3CBM9821,1234567890123456,200728121324,7,5,1142343069#		

58 Camera Sub-Stream Frame Interval:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command fromCCUto change thesub streamBit rate
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	114
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Frame Interval (1 to 100)	String	50
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$114,SU3339,DL3CBM9821,1234567890123456,20201120140846,7,45,3185509 390#		

59 Camera Video Recording Enable/Disable

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the camera Recording Status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	115
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Enable/Disable 1 Enable 0 Disable	String	0
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$115,SU3339,DL3CBM9821,123456789123456,20201120141119,7,1,4258865013#		

60 Camera Audio Recording ENABLE/DISABLE

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the camera Audio Record Status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	116
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
Packet Contents			
7	Camera no	String	1
8	Enable/Disable 1 Enable 0 Disable	String	1
Packet Footer:			
9	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$116,SU3339,DL3CBM9821,123456789123456,20201120141725,7,1,462952740#		

61 Camera Name for Display:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the camera name

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	117
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Camera Name	String	CAM1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$117,SU3339,DL3CBM9821,123456789123456,20201120142715,7,camera8888,34 67778330#		

62 Camera schedule-based/Continuous recording:

Frequency : Whenever OBU receives command from CCU.

Contents : Command from CCU to change the camera as schedule based/Continuous recording

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	118
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Continuous 0 Schedule based 1	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$118,SU3339,DL3CBM9821,123456789123456,20201120143120,1,1831480720#		

63 Camera schedule-based Recording Schedule:

Frequency : Whenever OBU receives command from CCU.

Contents : Command from CCU to change the camera as schedule based/Continuous recording

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	119
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Sunday (Enable 1, Disable 0)	String	0
8	Monday (Enable 1, Disable 0)	String	0
9	Tuesday (Enable 1, Disable 0)	String	0
10	Wednesday (Enable 1, Disable 0)	String	0
11	Thursday (Enable 1, Disable 0)	String	0
12	Friday (Enable 0, Disable 0)	String	0
13	Saturday (Enable 1, Disable 0)	String	0
14	StartTime (HHmm)	String	0600
15	EndTime (HHmm)	String	2200
Packet Footer:			
37	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
16	End of Packet	String	#
Sample packet :			
Example	\$119,SU3339,DL3CBM9821,123456789123456,20201120143422,1,1,1,1,1,1, 0900,1959,1357185077#		

64 Camera Video Expired Time:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the camera Video Expired Time

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	120
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	No of Days (0 to 255) 0 Overwritten in cyclic basis ☐	String	100
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$120, SU3339, DL3CBM9821, 123456789123456, 20201120143821, 50, 840383333#		

65 Camera Record Based on Motion Detection Enable/Disable:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to record Video based on motion detection

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	121
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Motion Detect (Enable/Disable) 1 Enable	String	1
9	Record Time (Min) (1 to 30)	String	3
Packet Footer:			
10	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
11	End of Packet	String	#
Sample packet :			
Example	\$121,SU3339,DL3CBM9821,123456789123456,20201218161907,1,1,3,331145 322 8#		

66 Camera Record Based on Motion Detection on sensitivity:

Frequency	: Whenever OBU receives command from CCU
Contents sensitivity	: Command from CCU to record Video based on motion detection on sensitivity
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	122
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Sensitivity (1 to 100)	String	10
9	Threshold (1 to 100)	String	50
10	Record Time (Min) (1 to 30)	String	3
Packet Footer:			
11	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
12	End of Packet	String	#
Sample packet :			
Example: \$122,SU3339,DL3CBM9821,123456789123456,20201218161914,1,10,50,3,3345714523#			

67 Camera Record Based on Motion Detection on Area:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to record the Video based on motion detection on Area Basis

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	123
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	X1	String	1
9	X2	String	10
10	Y1	String	2
11	Y2	String	13
12	Record Time (Min) (1 to 30)	String	3
Packet Footer:			
13	32-bit Checksum Insures No error in transmission (optional)	Long(String)	
14	End of Packet	String	#
Sample packet :			
Example:\$123,SU3339,DL3CBM9821,123456789123456,20201218161919,1,1,10,1,15,3,2514204516#			

68 Camera Pre-Event Record Time:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to record the video on pre-event time
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	124
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Record Time (Minutes) (1 to 30)	String	5
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$124,SU3339,DL3CBM9821,1234567890123456,20201120144744,2,922841634#		

69 Camera Post-Event Record Time:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to record the video on post-event time
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	125
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Record Time (Minutes) (1 to 30)	String	5
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$125,SU3339,DL3CBM9821,123456789123456,20201120144952,4,1031959289#		

70 Camera Resolution for Snapshot to send backend:

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to change the stream video resolution for snapshot/video when sending to backend server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	126
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Video Resolution Main Stream 0 Sub Stream ☐ Main ☐	String	1
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example: \$126,SU3339,DL3CBM9821,123456789123456,20201222144849,2,0,69072328#			

71 Frequency of sending Snapshot images to backend:

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set frequency to send images to backend server
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	127
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	Camera no	String	1
8	Frequency (min) (1 to 255)	String	5
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End of Packet	String	#
Sample packet :			
Example	\$127,SU3339,DL3CBM9821,1234567890123456,200728121324,1,4,1520913221#		

72 Set Ignition on/off Duration(CCU-OBU)

Frequency : Whenever OBU receives command from CCU.

Contents : Command from CCU to set the duration after ignition off to go into sleep mode/auto shutdown mode. Set frequency of data transmission to CCU in normal, sleep and emergency modes

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	128
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Sleep mode/shutdown Mode Sleep mode 0 Shutdown mode 1	String	1
8	Duration to Enter Sleep/Shutdown Mode (Min)	String	10
9	Frequency in sleep mode (min) (1 to 255)	String	5
10	Frequency in normal mode (seconds) (1 to 255)	String	5
Packet Footer:			
11	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
12	End of Packet	String	#
Sample packet :			
Example: \$128,SU3339,DL3CBM9821,1234567890123456,201222145022,1,10,15,5,3780884598#			

73 Get Firmware Version(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to send the Firmware version.
Whenever this command receives then OBU has to send Firmware version to server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	129
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$129,SU3339,DL3CBM9821,123456789123456,231122182512,0,3877744289#		

74 Get Protocol Version(CCUCU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to send the Protocol version.
Whenever this command receives then OBU has to send protocol version to server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	130
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	String
9	End of Packet	1	#
Sample packet :			
Example	\$130,SU3339,DL3CBM9821,123456789123456,201121094847,0,3492407914#		

75 Get Mac Address(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to send the Mac Address.
Whenever this command receives then OBU has to send Mac Address to server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	131
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example: \$131,SU3339,DL3CBM9821,123456789123456,201222145333,0,4214407592#			

76 Get Vehicle Registration Number(CCU-OBU)

Frequency : Whenever OBU receives command from CCU.

Contents : Command from CCU to send the Vehicle Registration Number Whenever this command receives then OBU Vehicle Registration Number to server.

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	143
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$143,SU3339,DL3CBM9821,123456789123456,201124100132,0,732927785#		

77 Get Ignition Mode(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to send the Vehicle Registration Number. Whenever this command receives then OBU Vehicle Registration Number to server has to send.

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	144
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$144,SU3339,DL3CBM9821,123456789123456,201121104747,0,1900373883#		

78 Get Primary IP Address(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to send the Primary IP Address.
Whenever this command receives then OBU has to send Primary IP Address to server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	132
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$132,SU3339,DL3CBM9821,123456789123456,201121105302,0,2583669216#		

79 Set Primary IP Address(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the Primary IP Address of server in OBU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	133
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	IP Address	String	222.168.32.1
Packet Footer:			
8	32 bit Checksum Insures No error in transmission(optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$133,SU3339,DL3CBM9821,123456789012345,201121105609,49.207.2.123,29 676 96432#		

80 Get Secondary IP Address(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to send the Secondary IP Address. Whenever this command receives then OBU has to send Secondary IP Address to server.
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	134
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32-bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$134,SU3339,DL3CBM9821,1234567890123456,201121105733,0,745002474#		

81 Set Secondary IP Address(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the Secondary IP Address in OBU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	135
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	IP Address	String	222.168.32.1
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$135,SU3339,DL3CBM9821,1234567890123456,201121105847,49.207.2.123,33 93164790#		

82 Set Port Number Primary IP(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set the Port Number of primary IP in OBU

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(3)	136
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Port number (integer with 2 bytes)	String	9999
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$136,SU3339,DL3CBM9821,123456789123456,201121110107,5557,1008019634#		

83 Set Port Number Secondary IP(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the Port Number of Secondary IP in OBU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(3)	137
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Port number (integer with 2 bytes)	String	9999
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$137,SU3339,DL3CBM9821,1234567890123456,201121110255,5558,1557839 427#		

84 Set Vehicle Registration Number(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set the vehicle Registration Number in OBU

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	138
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Vehicle Reg Number	String	TS10AP1234
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$138,SU3339,DL3CBM9821,123456789123456,201121110420,AP36TS1234,10 010 54310#		

85 Set APN(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set the APN in OBU

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	139
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	APN	String	internet
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$139,SU3339,DL3CBM9821,123456789123456,201222145516,AIRTEL GPRS.COM,857120790#		

86 Set Time Duration for Emergency State(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the Time duration for Emergency State
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	140
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Time Duration (Sec) (5 to 255)	String	10
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$140,SU3339,DL3CBM9821,1234567890123456,201222145559,15,2539748439#		

87 Get IMEI Number(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to send the IMEI Number. Whenever it receives this command send IMEI Number to CCU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	141
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
Packet Contents			
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$141,SU3339,DL3CBM9821,123456789123456,201121165250,0,3036730543#		

88 Transmission Mode Always ON(CCUCBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set transmission mode.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	142
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Emergency button Mode 1 Enable 0 Disable	String	0
8	SMS Mode 1 Enable 0 Disable	String	1
9	phone number	String	9898765432
Packet Footer:			
10	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
11	End of Packet	String	#
Transmission Mode Always ON Sample packet :			
Example	\$142,SU3339,DL3CBM9821,123456789123456,201121113504,1,0,9876545432,1830419193#		

89 System Reset (CCU to OBU)

Frequency : Event based

Contents : Everytime OBU receives SystemReset, OBU will restart the OBU

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	063
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End of Packet	String	#
Sample packet :			
Example	\$063,SU3339,DL3CBM9821,123456789123456,201121113549,0,1664423730#		

90 Permanent Schedule: (CCU to OBU)

Frequency :Event based ,every time the ccu wants to set Permament schedule,daily
this schedule will reload after 24 Hours in OBU

Contents : Permanent Schedule Packet (CCU to OBU)

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	061
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenumber where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Dateand Time, Format=YYMMDDHHMMSS	String	200728121324
7	Duty Id (Maximum16 character)Alphanumeric	String	1234
8	Dutystart DateTime Format: YYMMDDHHMMSS	String	200812122021
9	No of Trips	String	2
10	Route ID1 (6 Character)	String	20PUP
11	RouteID2	String	20PDOWN
		String	
	Route IDn (till no of trips)	String	
Packet Footer:			
n-1	32-bit ChecksumInsures No error in	Long(String)	
n	End Byte, Value="##"	String	#
Sample packet :			
Example	\$061,SU3339,DL3CBM9821,123456789123456,201222150002,123456,1201122145945,2,33,44,3725511406#		

91 Temporary Schedule (CCU to OBU)

Frequency : Event based, every time the CCU wants to set temporary Schedule

Contents : Temporary Schedule Packet (CCU to OBU)

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	062
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format = YYMMDDHHMMSS	String	200728121324
7	DutyId (Maximum 16 character) Alphanumeric	String	1234
8	Schedule start Date Time Format : YYMMDDHHMMSS	String	
9	Schedule End Date Time Format : YYMMDDHHMMSS	String	
10	No of Trips	String	1
11	Route ID1 (6 Character)	String	
12	Route ID2	String	
		String	
	Route ID n (till no of trips)	String	
Packet Footer			
n-1	32 bit Checksum Insures No error	Long (String)	
n	End Byte, Value = "#"	String	#
Sample packet :			
\$062,SU3339,DL3CBM9821,123456789123456,201222150137,123456,201222145945,201223145900,2,33,44,3055970357#			

92 Panic Messages(CCU-OBUS)

Frequency : Event based, triggered by CCUServer

Contents : Ad-Hoc message in English, once this message is received it announces and show the message on the display board

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	003
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Message ID	String	
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End Byte, Value = “#”	String	
Sample packet :			
Example	\$003,SU3339,DL3CBM9821,123456789123456,201121114525,3,3891412488#		

93 ADHOC Messages, in plain English text: (CCU to OBU)

Frequency	: Event based, triggered by CCUServer
Content	: Ad-Hoc message in English, once this message is received it announces and show the message on the display board
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	052
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Ad-Hoc Message Text, maximum length is 50 bytes	String	
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End Byte, Value = “#”	String	
Sample packet :			
Example	\$052,SU3339,DL3CBM9821,123456789123456,201121114635,jTextField1,270504 4856#		

94 File Download (CCU to OBU)

Frequency : Event based, everytime OBU receives File Download, OBU will download the file.

Contents : File Download Event

Channel : GPRS

Packet Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	060
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the	String	DL3CBM9821
5	15 Digit IMEInumber	String	123456789012345
6	DateandTime, Format=YYMMDDHHMMSS	String	200728121324
7	File Type 1Folder 0 File	String	1
8	File Name/folder name If folder not available givespace.	String(n)	Phonenum.txt
Packet Footer:			
9	32 bit ChecksumInsures No	Long(String)	
10	End Byte, Value="#"	String	#
Sample packet :			
Example	\$060,SU3339,DL3CBM9821,123456789123456,201121114800,0,123456,10564 789 13#		

95 Schedule Request Response (CCU to OBU)

Frequency : Event based, everytime CCUsend Schedule Request Response packet, whenever CCUreceives Schedule request packet from OBU.

Contents :Schedule Request Response

Channel : GPRS

Pacaket Details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character	String	\$
2	Header (FIXED 3	String (3)	064
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenunder	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMS S	String	200728121324
7	0/1 1 Success (Correct Duty ID) 0 Failure (Invalid Duty ID)	String	1
Packet Footer:			
8	32 bit Checksum	Long(String)	
9	End Byte, Value="#"	String	#
Sample packet :			
Example	\$064,SU333,DL3CBM9821,123456789012345,200728121324,1,10564789 13#		

96 Route Start (CCU to OBU)

Frequency : Event based, every time the OBU receives this message then OBU starts the specified route

Contents : Route-ID (CCU to OBU)

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	051
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenumber wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	String	String	200728121324
7	Route ID, Format = ABCDEF (Maximum 6character) Alphnumeric	String	ABCDEF
8	Start DateTime, Format =YYMMDDHHMMSS	String	120102111520
9	Expected Completion Time, Format =HHMM	String	1152
10	Duty Id, Format = 'X1Y2' or ' Y2'(Maximum 16 character)	String	1234
Packet Footer:			
11	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
12	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$051,SU3339,DL3CBM9821,123456789123456,201121115452,1,120102111520,1 152,123456,486424829#		

97 Route End (CCU to OBU)

Frequency : Event based, every time the OBU received this command if Route ID matches then OBU will end the route.

Content : Route-ID (CCU to OBU)

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	050
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Route ID, Format = ABCDEF (Maximum 6 character) Alphanumeric	String	ABCDEF
8	Start Date Time, Format =YYMMDDHHMMSS	String	
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
10	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$050,SU3339,DL3CBM9821,123456789123456,201121115605,1,201121115200,20 5444689#		

98 Request Health Packet(CCU-OBU)

Frequency : Event based, triggered by CCU Server

Contents : requesting health packet from OBU

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	065
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format= YYMMDDHHMMSS	String	200728121324
7	Reserved	String	
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$065,SU3339,DL3CBM9821,123456789123456,201121120055,0,708587102#		

99 Snapshot File Upload (CCU to OBU)

Frequency : Event based, everytime OBU receives File Upload, OBU will send the file to server at specified file location in FTP server

Contents : File Upload event

Channel : GPRS

Packet details :

S.no	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	070
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	From Date Time (YYMMDDHHMM)	String	
8	To Date Time (YYMMDDHHMM)	String	
9	Camera no Camera no (CAM1 1 ,CAM2 2 ,CAM3 3,CAM4 4 ,CAM5 5 ,CAM6 6 ,CAM7 7 ,CAM8 8)	String	1
10	Snapshot/Video Snapshot 0, Video 1	String	1
Packet Footer			
11	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
12	End Byte, Value="#"	String	#
Sample packet :			
Example	\$070,SU3339,DL3CBM9821,123456789123456,201121120526,201121120133,2 01121120133,5,0,3004573870#		

100 Common Acknowledgement to all packets(CCU to OBU)

Frequency : Event based, everytime CCU receives message from OBU, it sends back acknowledgement

Contents : Acknowledgment

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	ACK
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	1 ☐ success 0 ☐ Failure	String	1
8	Message type of Packet Current Route: 50 Ping Connection Packet: 90 NMEA: 4	String	4
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End Byte, Value=" #"	String	#
Sample packet :			
Example	\$ACK,SU3339,DL3CBM9821,123456789123456,200728121324,1,4,3004573870#		

101 SMS packet for Location(OBU-CCU)&(OBU-SMS)

Frequency : Event based, this packet will be sent to server and phone when GETLOCATION msg is received from phone

Contents : SMS Location Packet

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	ACTVR
3	OBUID	String(6)	000009
4	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
5	Firmware version	String	1.0.1
6	15 Digit IMEInumber	String	123456789012345
7	Alert ID	String	12
8	Latitudevalue in decimal degrees (with minimum6 decimal places)	double (String)	17.7823411
9	Latitude Direction(For example, N=North; S=South	String(1)	N
10	Longitude valuein decimal degrees (with minimum 6 decimal places)	double (string)	78.3123123
11	Longitude Direction (For example, E=East; W=West	String(1)	E
12	GPS Fix (1 = GPS fix OR 0 = GPS invalid)	String(1)	1
13	Date&Time ddmmyyyyhhmmss	String	16082022123342
14	Heading	String	263.19
15	Speed	String	25.4
16	GSM Strength	String	24
17	MCC	String	404
18	MNC	String	10
19	LAC Locationarea code	String	(214) (0x00D6)
20	Mainpower status	String	1
21	Ignition Status	String	1
22	Battery Voltage	String	24.6

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23	Framenumber	String	23333
24	Vehcile mode	String	ID
Packet Footer:			
25	End Byte, Value="#"	String	#
Sample Packets			
Example :\$ACTVR,000009,SU3339,10.0.11,123456789012345,11,17.47795000,N,78.57029500,E,1,20220720122344,328.0,0.0,0,404,73,D05,1,1,0.0,31,ID,#			

102 Video Loss

Frequency : Every time the Video is activated / deactivated, it is collected for a user defined interval and then send to server

Contents : Video no and status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	091
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEInumber	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Video No (1,2,3,4)		1
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	2757815055
9	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$091,SU3339,000009,123456789123456,201124144012,1,2757815055#		

103 Video Tamper

Frequency : Every time the Video is activated / deactivated, it is collected for a user defined interval and then send to server

Contents : Video no and status

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	092
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Video No (1,2,3,4)		1
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	2757815055
9	End Byte, Value = “#”	String	#
Sample packet :			
Example	\$092,SU3339,000009,123456789123456,201124144012,1,2757815055#		

104 SET APN SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set APN.

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	SETAPN
Packet Contents:			
3	APN	String	sensem2m2
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETAPN,sensem2m2#			

105 SET IP1 SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket to set ipaddress .

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	SETIP1
Packet Contents:			
3	Ipaddress1	String	192.168.1.1
4	Port1	String	2222
Packet Footer:			
5	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETIP1,192.168.1.1,2222#			

106 SET IP2 SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set iPAddress .

Channel : SMS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	SETIP2
Packet Contents:			
3	IpAddress2	String	192.168.1.2
4	Port2	String	2223
Packet Footer:			
5	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETIP2,192.168.1.2,2223#			

107 SET VEHICLE REGISTRATION NUMBER SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set Vehicle

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	SETVEHREG
Packet Contents:			
3	Vehicle Registration number	String	TS0909
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETVEHREG,TS0909#			

108 SET IGNITION ON/OFF DURATION SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set ignition on/off status

Channel : SMS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	SETIGNCFG
Packet Contents:			
3	Ignition on/off duration	String	2
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETIGNCFG,2#			

109 SET EMERGENCY DURATION SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set emergency state duration

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	SETEMRCFG
Packet Contents:			
3	Emergency state Duration	String	5
Packet Footer:			
4	End Byte, Value=" #"	String	#
Sample Packets			
Example : \$SETEMRCFG,5#			

110 RESET DEVICE SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to reset device

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	SETDEVRST
Packet Footer:			
3	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETDEVRST#			

111 GET IMEI NUMBER SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get IMEI number.when GETIMEI is given.

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (7)	GETIMEI
Packet Contents:			
3	IMEI number	String	123456789012345
Packet Footer:			
5	End Byte, Value="#"	String	#
Sample Packets			
Example :\$GETIMEI,123456789012345#			

112 GET APN SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get APN. When “GETAPN” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	GETAPN
Packet Contents:			
3	APN	String	sensem2m2
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETAPN,sensem2m2#			

122 GET IPADDRESS1 SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toget ipaddress . When “GETIP1” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	GETIP1
Packet Contents:			
3	Ipaddress1	String	192.168.1.1
4	Port1	String	2222
Packet Footer:			
5	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETIP1,192.168.1.1,2222#			

114 GET IPADDRESS2 SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get iPaddress . When “GETIP2” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (6)	GETIP2
Packet Contents:			
3	Ipaddress2	String	192.168.1.2
4	Port2	String	2223
Packet Footer:			
5	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETIP2,192.168.1.2,2223#			

115 GET VEHICLE REGISTRATION NUMBER SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get Vehicle Reg no. When “GETVEHREG” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	GETVEHREG
Packet Contents:			
3	Vehicle Registration number	String	TS0909
Packet Footer:			
4	EndByte, Value=“#”	String	#
Sample Packets			
Example : \$GETVEHREG,TS0909#			

116 GET IGNITION ON/OFF DURATION SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get ignition on/off status. When “GETIGNCFG” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	GETIGNCFG
Packet Contents:			
3	Ignition on/off duration	String	2
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETIGNCFG,2#			

117 GET EMERGENCY DURATION SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get emergency state duration. When “GETEMRCFG” is given from phone

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	GETEMRCFG
Packet Contents:			
3	Emergency state Duration	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETEMRCFG,5#			

118 SMS Health packet(OBU-CCU)(OBU-SMS)

Frequency : Event based, this packet will be sent to server and phone when GETHEALTH packet is received from phone

Contents : SMS HealthLocation Packet

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	HCHKR
3	OBUID	String(6)	000009
4	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
5	Firmware version	String	1.0.1
6	15 Digit IMEInumber	String	123456789012345
7	Alert ID	String	12
8	Latitude value in decimal degrees (with minimum 6 decimal places)	double (String)	17.7823411
9	Latitude Direction (For example, N=North; S=South)	String(1)	N
10	Longitude value in decimal degrees (with minimum 6 decimal places)	double (string)	78.3123123
11	Longitude Direction (For example, E=East; W=West)	String(1)	E
12	GPS Fix (1 = GPS fix OR 0 = GPS invalid)	String(1)	1
13	Date&Time ddmmyyyyhhmmss	String	16082022123342
14	Heading	String	263.19
15	Speed	String	25.4
16	GSM Strength	String	24
17	MCC	String	404
18	MNC	String	10
19	LAC Location area code	Short (String)	(214) (0x00D6)
20	Main power status	String	1
21	Ignition Status	String	1
22	Battery Voltage	String	24.6
23	Frame number	String	23333
24	Vehcile mode	String	ID
Packet Footer:			
25	End Byte, Value="#"	String	#
Sample Packets			
Example: \$HCHKR,000009,SU3339,10.0.11,123456789012345,11,17.47795000,N,78.57029500,E,1,20220720122344,328.0,0.0,0,404,73,D05,1,1,0.0,31,ID,#			

119 SET SMS CLEAR Default configuration

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket to set Default configuration

Channel : SMS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (9)	CLRCGFALL
Packet Footer:			
3	End Byte, Value="#"	String	#
Sample Packets			
Example : \$CLRCFGALL#			

120 SET CLEAR OVERSPEED BUZZER alarm SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket to set over speed buzzer disable

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(10)	CLRALMSPEED
Packet Footer:			
3	End Byte, Value="#"	String	#
Sample Packets			
Example : \$CLRALMSPEED#			

121 SET PHONE NUMBER SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to set phone number

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	SETPHNUM
3	Phone number	String	9989876511
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETPHNUM,9989876511#			

122 GET OVERSPEED SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to get OVERSPEEDLIMIT, when “GETOSLIMIT” is given from phone

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(10)	GETOSLIMIT
Packet Contents:			
3	OVER SPEED LIMIT	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETOSLIMIT,5#			

123 GET HARSH ACCELERATION SMS

Frequency	: Event based, when GPRS disconnected for configured interval
Contents	: SMS Packet to get harshacceleration. when “GETHA” is given from phone
Channel	: SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	GETHA
Packet Contents:			
3	Harshacceleration Duration	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETHA,5#			

124 GET HARSH BREAK SMS

Frequency	: Event based, when GPRS disconnected for configured interval
Contents	: SMS Packet to get harsh BREAK. . when “GETHB” is given from phone
Channel	: SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	GETHB
Packet Contents:			
3	Harshbreak	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETHB,5#			

125 GET RASH TURN SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toget rashturn. . when “GETRT” is given from phone

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	GETRT
Packet Contents:			
3	Harshacceleration Duration	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$GETRT,5#			

126 SET OVERSPEED SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMS Packet to Set OVERSPEED LIMIT

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String(10)	SETOSLIMIT
Packet Contents:			
3	OVER SPEED LIMIT	String	5
Packet Footer:			
4	End Byte, Value=" #"	String	#
Sample Packets			
Example : \$SETOSLIMIT,5#			

127 SET HARSH ACCELERATION SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toSet harsh acceleration

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	SETHA
Packet Contents:			
3	Harshacceleration Duration	String	5
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example :\$SETHA,5#			

128 SET HARSH BREAK SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toSet harsh BREAK

Channel : SMS

Packet details:

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	SETHB
Packet Contents:			
3	Harshbreak	String	5
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETHB,5#			

129 SET RASH TURN SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toSet rash turn

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (5)	SETRT
Packet Contents:			
3	RASH TURN Duration	String	5
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETRT,5#			

130 CLEAR command to set default settings (CCU TO OBU)

Frequency : Event based, triggered by CCUServer

Contents :SET Default configuration

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	072
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Message Text, maximum length is 50 bytes	String	
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End Byte, Value = “#”	String	
Sample packet :			
Example	\$072,SU3339,DL3CBM9821,1234567890123456,201121114635,270504 4856#		

131 CLEAR COMMAND TO SET OVERSPEED BUZZER DISABLE (CCU TO OBU)

Frequency : Event based, triggered by CCU Server

Contents : Set Default Over speed Buzzer Disable

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	071
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents:			
4	Vehiclenumber wherethe device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	MessageText, maximum length is 50 bytes	String	
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long (String)	
9	End Byte, Value = “#”	String	
Sample packet :			
Example	\$071,SU3339,DL3CBM9821,123456789123456,201121114635,270504 4856#		

132 Get APN(CCUCU-OBUCU)

Frequency : Whenever OBUCU receives command from CCUCU

Contents : Command from CCUCU to send the APN
Whenever this command receives then OBUCU has to send APN to server

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	145
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethe device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Data	String	0
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$145, SU3339,DL3CBM9821,123456789123456,201121105302,0,2583669216#		

133 Send APN(OBU-CCU)

Frequency : Event Based

Contents : Whenever OBU receives **Get APN Address** Command,OBU has to send this packet to ccu

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String (1)	\$
2	Header (FIXED 3 chars)	String (3)	146
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format=YYMMDDHHMMSS	String	200728121324
7	If getapn came through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	APN	String	SENSEM2M2
Packet Footer:			
9	32 bit ChecksumInsures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$146,3339,AP36TS1234,864819050843380,240312115822,49.207.2.123,2255,jionet,53fd0469#		

134 Set HARSH ACCELERATION(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the harsh acceleration of server in OBU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	079
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Harsh acceleration	String	2
Packet Footer:			
8	32 bit Checksum Insures No error in transmission(optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$079,SU3339,DL3CBM9821,123456789012345,201121105609,2,29676 96432#		

135 Set HARSH BREAK(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set the harsh BREAK of server in OBU

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	080
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Harsh break	String	2
Packet Footer:			
8	32 bit Checksum Insures No error in transmission(optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$080,SU3339,DL3CBM9821,123456789123456,201121105609,2,29676 96432#		

136 Set RASH TURN(CCU-OBU)

Frequency : Whenever OBU receives command from CCU

Contents : Command from CCU to set the rash turn of server in OBU

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	082
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	RASH TURN	String	26
Packet Footer:			
8	32 bit Checksum Insures No error in transmission(optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$082,SU3339,DL3CBM9821,123456789123456,201121105609,26,29676 96432#		

137 Set OVERSPEED(CCU-OBU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from CCU to set the OVERSPEED of server in OBU
Channel	: GPRS
Packet details	:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	074
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents:			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	OVER SPEED	String	26
Packet Footer:			
8	32 bit Checksum Insures No error in transmission(optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$074,SU3339,DL3CBM9821,1234567890123456,201121105609,26,29676 96432#		

138 Get HARSH ACCELERATION(CCU-OBU)

Frequency : Event Based

Contents : Command from CCU to OBU for getting harshacceleration value

Channel : GPRS

Packet details :

S n o	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	076
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Harsh acceleration	String	20
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$076,SU3339,000009,123456789123456,201124114125,20,1460835457#		

139 GET HARSH BREAK(CCU-OBU)

Frequency : Event Based

Contents : : Command from CCU to OBU for getting harsh break value

this packet

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	077
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	Harsh BREAK	String	20
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$077,SU3339,000009,123456789123456,201124114125,20,1460835457#		

140 GET RASH TURN(CCU-OBU)

Frequency : Event Based

Contents :: Command from CCU to OBU for getting rash turn value

Channel : GPRS

Packet details:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	078
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	RASHTURN	String	20
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$078,SU3339,000009,123456789123456,201124114125,20,1460835457#		

141 GET OVER SPEED(CCU-OBU)

Frequency : Event Based

Contents : Command from CCU to OBU for getting overspeed value

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	075
3	Vendor ID (FIXED 6 chars)	String(6)	SU3339
Packet Contents			
4	Vehicle number where the device was	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	OVERSPEED	String	45
Packet Footer:			
8	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
9	End of Packet	String	#
Sample packet :			
Example	\$075,SU3339,000009,123456789123456,201124114125,987987987,45,1460835457#		

142 Send HARSH ACCELERATION(OBU-CCU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from OBU to CCU sending harsh acceleration value
Channel	: GPRS
Packet details	:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	076
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	If getha came through Server/SMS Server-ip address and port SMS-phonenum	String	9898989898
8	Data	String	0
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$076,3339,AP36TS1234,864819050843380,240312115536,49.207.2.123,2255,20,2679a7f3#		

143 Send HARSH BREAK(OBU-CCU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from OBU to CCU sending harsh break value
Channel	: GPRS
Packet details	:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	077
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If gethb came through Server/SMS Server-ip address and port SMS-phonenum	String	9898989898
8	Data	String	0
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$077,3339,AP36TS1234,864819050843380,240312115554,49.207.2.123,2255,20,1ee6be68#		

144 Send RASH TURN(OBU-CCU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from OBU to CCU sending rash turn value
Channel	: GPRS
Packet details	:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	078
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If getrt came through Server/SMS Server-ip address and port SMS-phonenummer	String	9898989898
8	Data	String	0
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$078,3339,AP36TS1234,864819050843380,240312115609,49.207.2.123,2255,60,207150c3#		

145 Send OVER SPEED(OBU-CCU)

Frequency	: Whenever OBU receives command from CCU
Contents	: Command from OBU to CCU sending overspeed value
Channel	: GPRS
Packet details	:

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	075
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format=YYMMDDHHMMSS	String	200728121324
7	If getos came through Server/SMS Server-ip address and port SMS-phonenum	String	9898989898
8	Data	String	0
Packet Footer:			
9	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
10	End of Packet	String	#
Sample packet :			
Example	\$075,3339,AP36TS1234,864819050843380,240312115625,49.207.2.123,2255,45,32e86d2b#		

146 CAN Data(OBU-CCU)

Contents : Can information including all can parameters will be sent to server .

Channel : GPRS

CommunicationDetails : When GPRS is connected CAN data is sent to server by specified interval

Packet Details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	014
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	DateandTime, Format=ddmmyyyy	String	120322
7	Timestamp(HH:mm:ss)	String	123012
8	CAN1 Parameter	String	CAN1:
9	Odometer Reading	String	40543.23
10	Vehicle Status (0/1)	String	0.0
11	SoC (State of Charge)	String	60.93
12	SoH (State of Health)	String	100.0
13	Speed	String	0.0
14	Acceleration	String	0.0
15	Voltage Pack Level	String	24.55
16	Current Pack Level	String	12
17	Charger Current	String	670.0
18	Charger Energy/Voltage	String	119.0
19	Ambient Temperature	String	50.22
20	Cell Temperature	String	60.16
21	Motor Temperature	String	50
22	BMS Error	String	3.0
23	Gear Position R	String	0.0
24	Gear Position N	String	0.0
25	Gear Position D	String	0.0
26	Front Door (0/1)	String	0.0
27	Middle Door (0/1)	String	0.0
28	CAN2 Parameter	String	CAN2:
29	Odometer Reading	String	0.0
30	Vehicle Status (0/1)	String	1.0
31	SoC (State of Charge)	String	0.0
32	SoH (State of Health)	String	0.0
33	Speed	String	40.3
34	Acceleration	String	20.99
35	Voltage Pack Level	String	0.0
36	Current Pack Level	String	0.0
37	Charger Current	String	0.0

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38	Charger Energy/Voltage	String	0.0
39	Ambient Temperature	String	0.0
40	Cell Temperature	String	0.0
41	Motor Temperature	String	0.0
42	BMS Error	String	0.0
43	Gear Position R	String	1.0
44	Gear Position N	String	1.0
45	Gear Position D	String	1.0
46	Front Door (0/1)	String	1.0
47	Middle Door (0/1)	String	1.0
Packet Footer:			
48	32 bit Checksum Insures No error in transmission (optional)	Long(String)	d5e77a72
49	End of Packet	String	#
Sample packet :			
Example	\$014,3339,DL1PD8365,863418058197328,24102025,175758,CAN1:;5270.6,0.0,92.0,100.0,0.0,0.0,23.0,28.0,690.0,118.4,26.0,40.0,30.0,1.0,0.0,0.0,0.0,0.0,0.0,CAN2:;0.0,1.0,0.0,0.0,30.0,1.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,1.0,1.0,1.0,68b0fa4b#		

147 APC Packet On Every Stop(OBU-CCU)

Contents : APC information including passenger in and out count for both the doors will be sent to server .

Channel : GPRS

Communication Details : When GPRS is connected APC data is sent to server After five meters away from each stop.

Packet Details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	009
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethe device	String	DL3CBM9821
5	Packet Status L=Live or H= History	String(1)	L
6	15 Digit IMEI number	String	123456789012345
7	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
8	Latitude value in decimal degrees (with minimum 6 decimal places)	double(String)	17.7823411
9	Latitude Direction (For example, N=North;S=South)	String(1)	N
10	Longitude value in decimal degrees (with minimum 6 decimal places)	double(string)	78.3123123
11	Longitude Direction (For example, E=East;W=West)	String(1)	E
12	Stop Name	String	RGI Airport
13	Route Number	String	13U
14	Front APC IN	String	12
15	Front APC OUT	String	13
16	Back APC IN	String	14
17	Back APC OUT	String	15
Packet Footer:			
18	32 bit Checksum Insures No error in transmission (optional)	Long(String)	
19	End of Packet	String	#
Sample packet :			
Example	\$009, SU3339, DL3CBM9821,L, 123456789012345, 200728121324, 17.7823411, N, 78.3123123, E, RGI Airport, 13U,12,13,14,15, 2583669216#		

148 APC Packet on Route End(OBU-CCU)

Contents : APC information including passenger in and out count for both the doors will be sent to server .

Channel : GPRS

CommunicationDetails : When GPRS is connected APC data is sent to server on route end.

Packet Details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	012
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	Packet Status L=Live or H= History	String(1)	L
6	15 Digit IMEI number	String	123456789012345
7	Date, Format= YYMMDD	String	200728
8	Start Time Format=HHMMSS	String	101530
9	End Time Format=HHMMSS	String	123010
10	Route Name	String	ECIL TO SEC
12	Route Number	String	13U
13	Front APC IN	String	12
14	Front APC OUT	String	13
15	Back APC IN	String	14
16	Back APC OUT	String	15
Packet Footer:			
17	32 bit ChecksumInsures No errorin transmission (optional)	Long(String)	
18	End of Packet	String	#
Sample packet :			
Example	\$012,3339,AP36TS1234,L,864819050843380,240312,101530,123010, ECIL TO SEC,12C3,8,3,5,12,382b6bcf#		

149 Conductor Duty Start Packet(OBU to CCU) :

Frequency : Event based, everytime when conductor dutystarts.

Contents : Conductor Dutyidfor the respective duty.

Channel : GPRS

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	147
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234
8	Reserved String	String	
Packet Footer:			
9	32 bit Checksum Insures No error in transmission	Long(String)	
10	End Byte, Value="#"	String	#
Sample packet :			
Example	\$147,SU3339,000009,123456789123456,201123160206,1234,0,3609845968#		

150 Conductor Duty End Packet (OBU to CCU) :

Frequency : Event based, everytime when conductor duty ends.

Contents : Conductor Dutyid for therespectiveduty.

Channel : GPRS

Packet details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	147
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number where the device was installed	String	DL3CBM9821
5	15 Digit IMEI number	String	123456789012345
6	Date and Time, Format= YYMMDDHHMMSS	String	200728121324
7	Duty id	String	1234
8	Reserved String	String	
Packet Footer:			
9	32 bit Checksum Insures No error in transmission	Long (String)	
10	End Byte, Value=“#”	String	#
Sample packet :			
Example	\$148,SU3339,000009,123456789123456,201123160206,1234,0,3609845968#		

151 Alert on Emergency button press to phone through sms(OBU-Phone)

Frequency : Onpress of Emergency Button

Contents : Emergency Message on Press of Emergency button this packet is sent to phone

Channel : GPRS

Communication Details : When GPRS is connected, Send emergency message to dual IP Address. Until OBU receives STP message from server send the packet with configurable interval of time. Once OBU receives STP message send STOP_ACK message to server

Packet details :

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String(1)	\$
2	Packet Header	String (3)	EPB
3	Message type(FIXED 3 chars)	String(3)	EMR/SEM
4	IMEI(Vehicleid)	String	123456789012345
5	Packet Type	String	NM / SP
6	Date and time(ddmmyyyyhhmmss format)	String	20200728122035
7	GPS valid (A-Valid, V-Invalid)	String	A
8	Latitude in decimal degrees - dd.mmmmmm format	String	17.1234561231
9	Latitude Direction N – North; S – South	String	N/S
10	Longitude in decimal degrees - dd.mmmmmm format	String	78.123456781
11	Longitude Direction E – East; W – West	String	E/W
12	Altitude in ‘m’	String	235.123312312
13	Speed in km/h	String	38.123
14	Distance calculated from previous GPS data	String	123.31
15	Provider (G – FineGPS N – Coarse GPS or data from the network)	String	G/N
16	Vehicle Registration number	String	AP361234
17	Reply Number (The mobile number to which test response need to be sent)	String	9876543210
18	LAC Location area code	Short (String)	(214) (0x00D6)
19	Cell ID GSM Cell ID	(String)	53181 (0xCFBD)
Packet Footer:			
20	End Of Packet	String	#
Sample Packet			
Example	\$EPB,EMR,123456789012345,NM,20022024123324,A,17.1234561231,N,78.123456781,E,235.123312312, 38.123,123.31,G/N,AP361234,9876543210,234,AACD#		

152 GET PHONE NUMBER SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toget phone number. When GETPHNUM is given from phone.

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	GETPHNUM
3	Phone number	String	9989876511
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$GETPHNUM,9989876511#			

153 SET NORMAL TIME SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket toset normal time.

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	SETNORMALTIME
3	time	String	10
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$SETNORMALTIME,10#			

154 GET NORMAL TIME SMS

Frequency	: Event based, when GPRS disconnected for configured interval
Contents	: SMS Packet to get normal time. When “GETNORMALTIME” is given from phone
Channel	: SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	GETNORMALTIME
3	time	String	10
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example :\$ GETNORMALTIME,10#			

155 GET IGNITIONDEALYCONFIG TIME SMS

- Frequency** : Event based, when GPRS disconnected for configured interval
- Contents** : SMS Packet to get ignition delay time. When “GETIGNDELAYCFG” is given from phone.
- Channel** : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	GETIGNDELAYCFG
3	time	String	5
Packet Footer:			
4	End Byte, Value=“#”	String	#
Sample Packets			
Example : \$ GETIGNDELAYCFG,5#			

156 SET IGNITIONDEALYCONFIG TIME SMS

Frequency : Event based, when GPRS disconnected for configured interval

Contents : SMSPacket to set ignition off time.

Channel : SMS

Packet details:

Sno	Contents	No of Bytes	Sample Data
Packet Header:			
1	Start Character (FIXED 1 char)	String	\$
2	Header	String(8)	SETIGNDELAYCFG
3	time	String	5
Packet Footer:			
4	End Byte, Value="#"	String	#
Sample Packets			
Example : \$ SETIGNDELAYCFG,5#			

159 ALCOBRAKE

Frequency : On blow air through ALCOBRAKE device

Contents : Pass/Fail on blow air through ALCOBRAKE device

Channel : GPRS

Packet details :

Sno	Contents	No. of Bytes	Sample data
Packet Header:			
1	Start Character(FIXED 1 char)	String	\$
2	Header(FIXED 3 char)	String	015
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
3	Vehicle number where the device was installed	String	DL3CBM9821
4	15 Digit IMEI number	String	123456789012345
5	Date ddmmyyyyhhmmss	String	20122024123314
6	Result (0 for Fail/1 for Pass)	St	1
Packet Footer			
7	32 bit ChecksumInsures No error in transmission	Long(String)	
8	End Character (FIXED 1 char)	String	#
Sample Packet			
Example	\$015,SU3339,DL3CBM9821,123456789012345,22012025143101,1, 3609845968#		

160 APC Packet On Day End(OBU-CCU)

Contents : APC information including passenger in and out count for both the doors will be sent to server .

Channel : GPRS

Communication Details : When GPRS is connected APC data is sent to server at Day End.

Packet Details :

S.No	Contents	No of Bytes	Sample Data
Packet Header			
1	Start Character (FIXED 1 char)	String	\$
2	Header (FIXED 3 chars)	String (3)	013
3	Vendor ID (FIXED 6 chars)	String (6)	SU3339
Packet Contents			
4	Vehicle number wherethedevic	String	DL3CBM9821
5	Packet Status L=Live or H= History	String(1)	L
6	15 Digit IMEI number	String	123456789012345
7	Date, Format= YYMMDD	String	200728
8	End Time Format=HHMMSS	String	123010
9	Total APC IN	String	150
10	Total APC OUT	String	151
Packet Footer:			
11	32 bit ChecksumInsures No errorin transmission (optional)	Long(String)	
12	End ofPacket	String	#
Sample packet :			
Example	\$013,3339,AP36TS1234,L,864819050843380,240312,101530,150,151,382b6bcf#		